

GOOD MORNING!

早上好!

안녕하세요!

DAY 6

DAY 5 (FINAL PROJECT)

- 비즈니스/System 요구 사항 업데이트
- 시스템 설계 및 프로세스 정립
 - System(High Level) Design
 - Schedule/Time Management
- 역할 분담 및 일정 조율
- 개발 환경 구축(맵 디자인, SW 개발, 문서 통합 관리)
- 멀티 로봇 환경 구축 및 네비게이션
- 멀티 로봇 개별 업무 수행
- 멀티 로봇 협동 업무 수행
- (Optional) Turtlebot4 각종 센서 데이터의 이해와 적용

DAY 6 (FINAL PROJECT)

- 시스템 설계에 기반한 객체 감지 모델 구현
 - 로봇 환경에 적용 및 Unit Test
 - 모듈로 제작하고 launch파일로 구현
 - code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑
- 시스템 설계에 기반한 SysMon 설계 구현
 - 로봇 환경에 적용 및 Unit Test
 - 모듈로 제작하고 launch파일로 구현
 - code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 7 (FINAL PROJECT)

- 시스템 설계에 기반한 **AMR 제어** 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch 파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

프로젝트 RULE NUMBER ONE!!!

Have Fun Fun Fun!



PROJECT SCHEDULE **UPDATE** BY EACH TEAM



BUSINESS/SOLUTION REQUIREMENT **UPDATE** BY EACH TEAM

Using the posted notes and flipchart as needed

SYSTEM REQUIREMENT **UPDATE** BY EACH TEAM

Using the posted notes and flipchart as needed

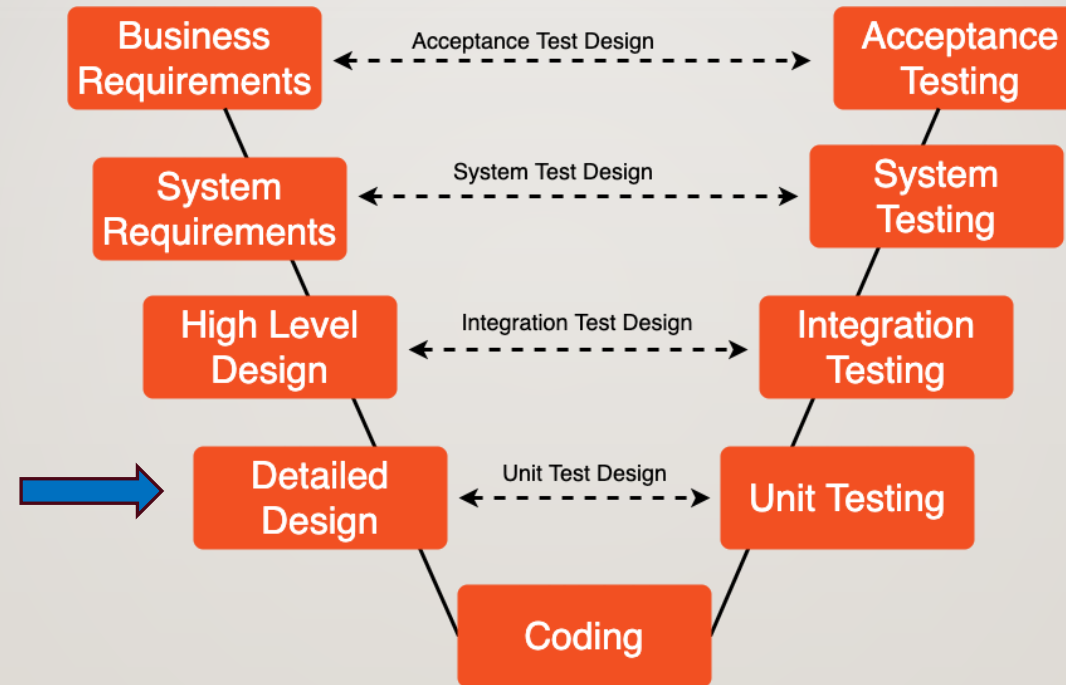
SYSTEM DESIGN ARCHITECTURE **UPDATE** BY EACH TEAM

Using the posted notes and flipchart as needed

SYSTEM DESIGN ARCHITECTURE
PRESENTATION SHOULD BE DONE HERE
BUT POSTPONED TO **DAY 7 AFTERNOON!!!!**



SPRINT I

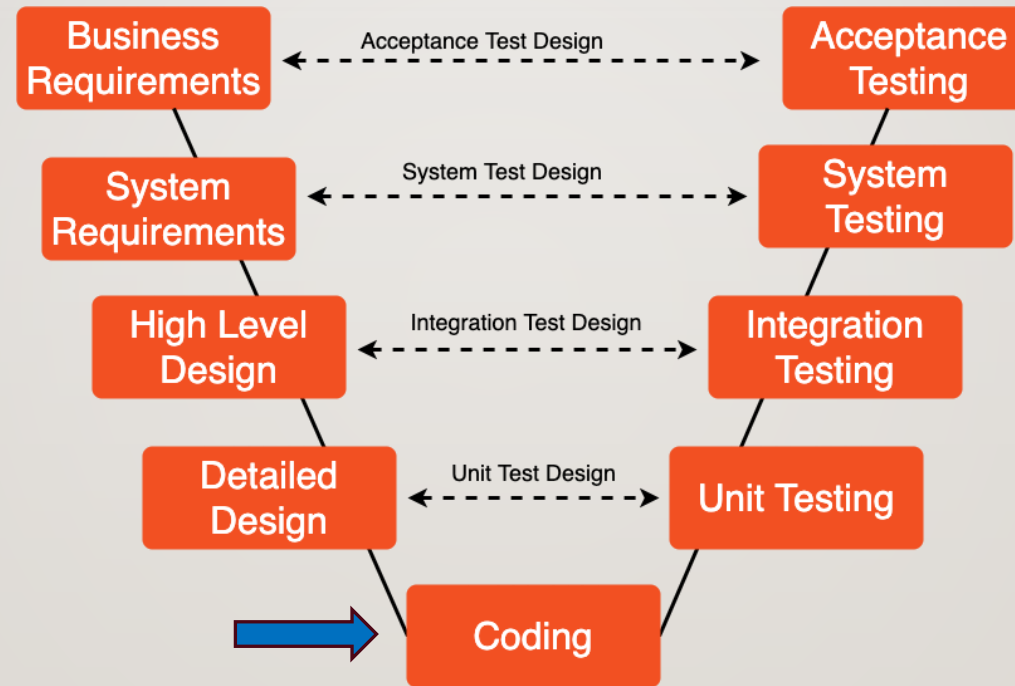


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TEAM EXERCISE 12

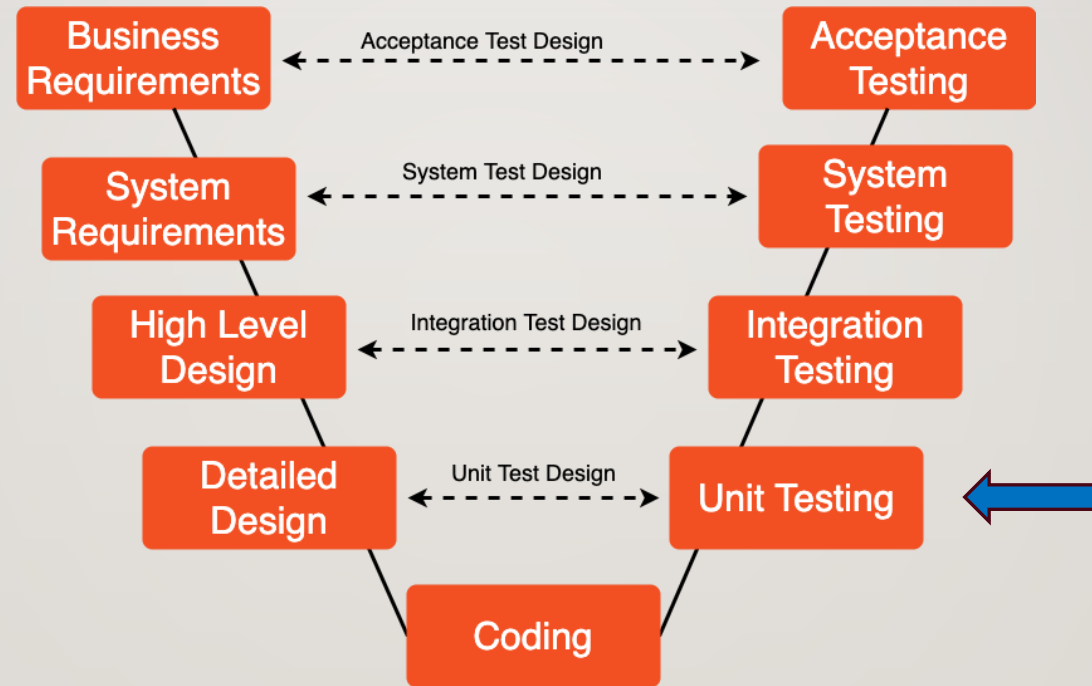
Perform Detail Design of Sprint I using Process Flow Diagram

SPRINT I



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SPRINT I



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TEAM EXERCISE 13

Perform coding and testing of Sprint 1 Module

(평가 2) DESIGN/CODE/TEST REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

WHAT TO DO IN CODE REVIEW

- Explain
 - Design

Topic: sensor_msg:

Type: xxxx

Data_a: {int} # number of detection

Data_b: {list} # contains list of class types

...

PGM.py

Node/Class_A:

var_a = xxx

var_b = yyy

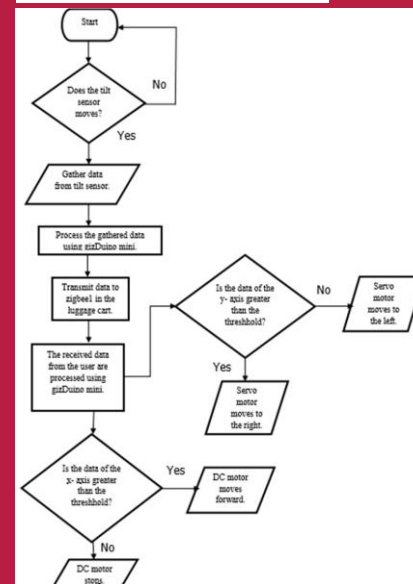
....

Main():

Call Func_A;

....

Func_A:



Topic: out_msg:

Type: xxxx

Data_a: {Bool} # True or False

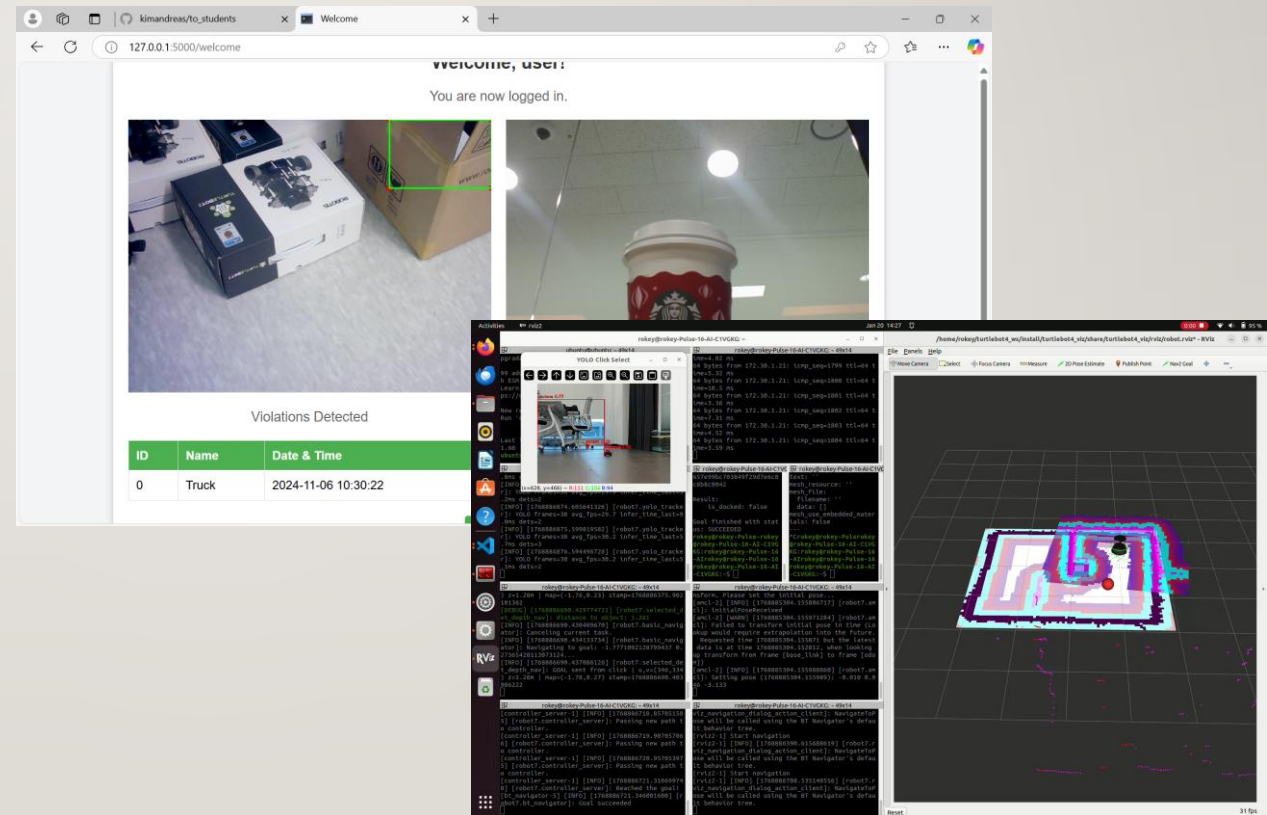
WHAT TO DO IN CODE REVIEW

- Explain
 - Design
 - Code

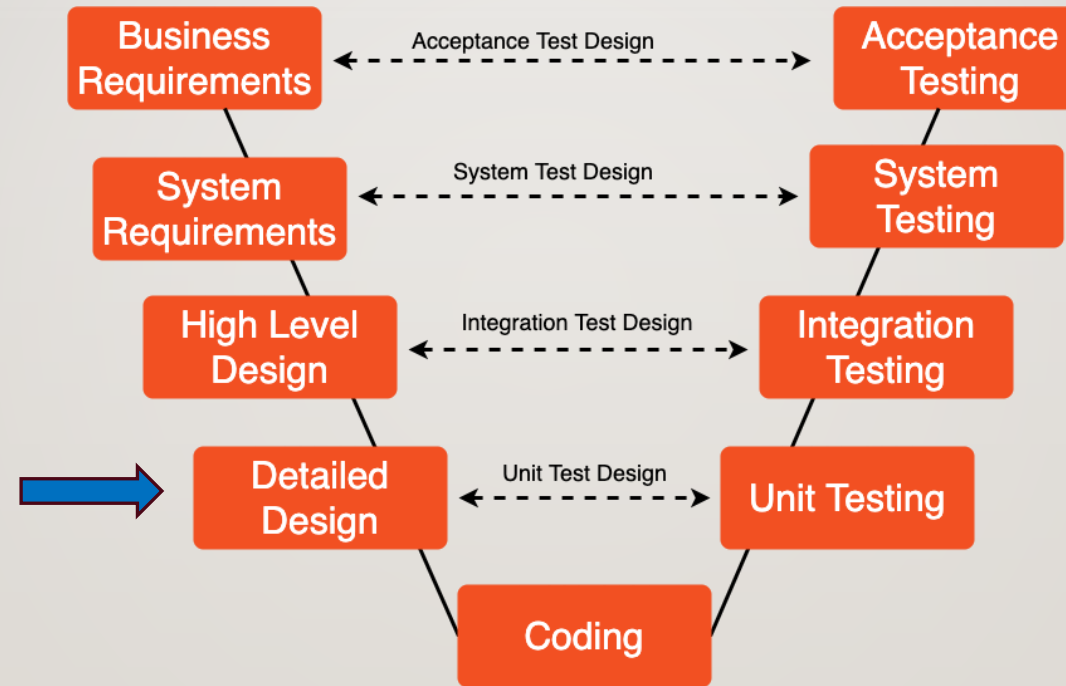
```
src > day3 > day3 > 3_1_c_follow_waypoints.py > main
12 # distributed under the
13 # WITHOUT WARRANTIES OR
14 # See the License for the specific language governing permissions and
15 # limitations under the License.
16 #
17 # @author Roni Kreinin (rkreinin@clearpathrobotics.com)
18
19 import rclpy
20
21 from turtlebot4_navigation.turtlebot4_navigator import TurtleBot4Directions, TurtleBot4Navigator
22
23 def main():
24     rclpy.init()
25     (variable) navigator: TurtleBot4Navigator
26     navigator = TurtleBot4Navigator()
27
28     # Start on dock
29     if not navigator.getDockedStatus():
30         navigator.info('Docking before intialising pose')
31         navigator.dock()
32
33     # Set initial pose
34     initial_pose = navigator.getPoseStamped([0.0, 0.0], TurtleBot4Direction.NORT
35     navigator.setInitialPose(initial_pose)
36
37     # Wait for Nav2
38     navigator.waitUntilNav2Active()
39
40 # Set goal poses
```


WHAT TO DO IN CODE REVIEW

- Explain
 - Design
 - Code
- Test/Validation
 - Test Code
 - Log Output
 - Screenshot/video
 - Photo/Video
 -



SPRINT 2

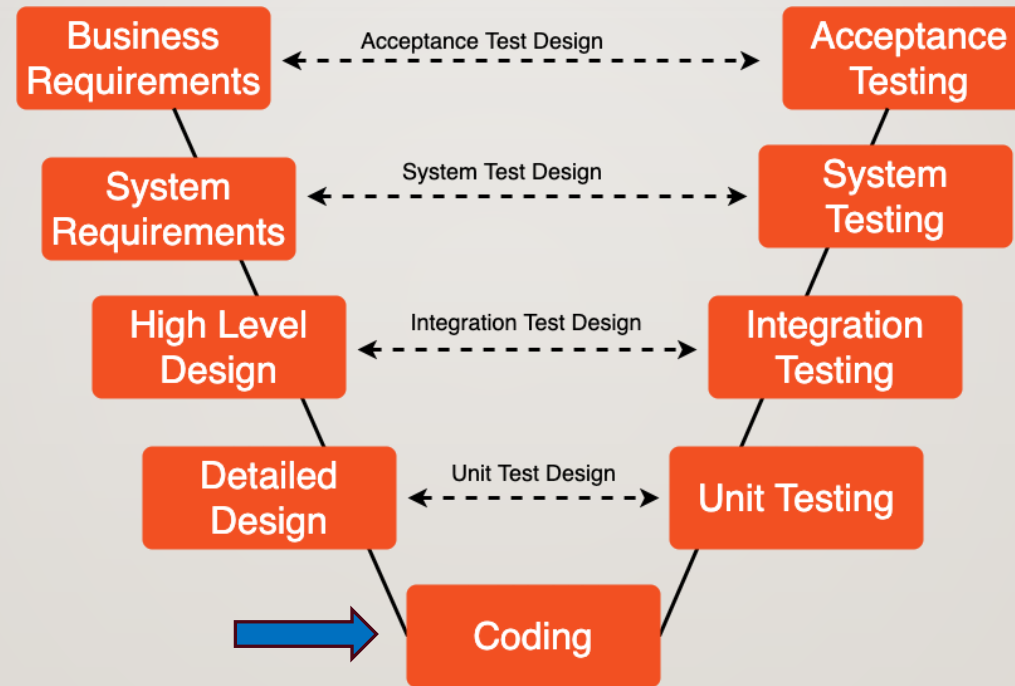


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TEAM EXERCISE 14

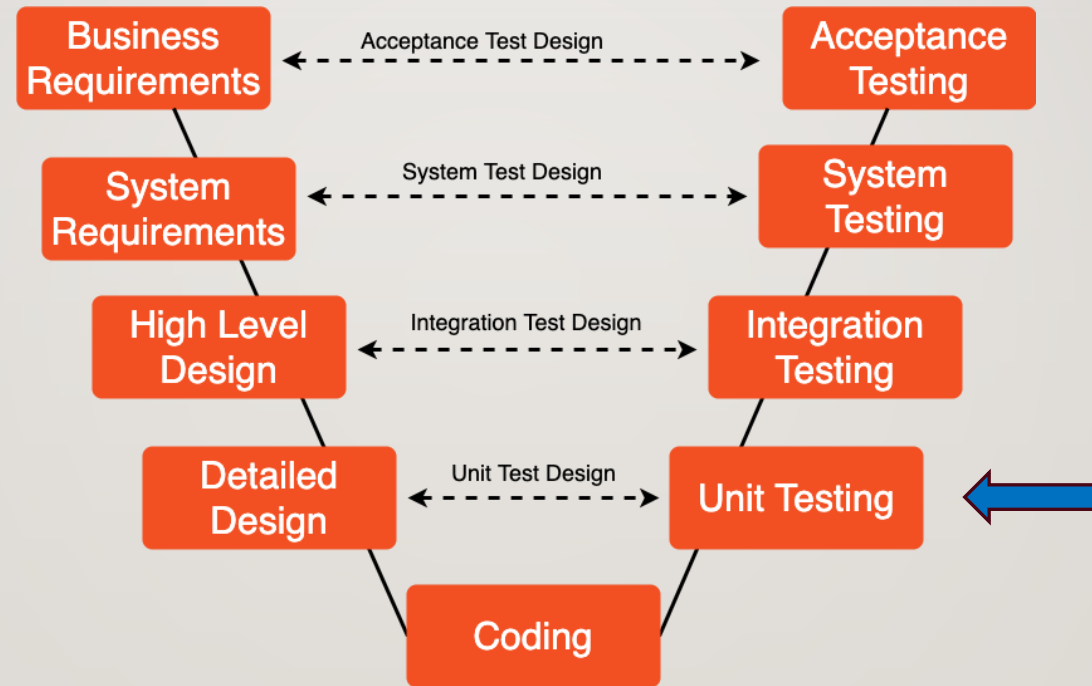
Perform Detail Design of Sprint 2 using Process Flow Diagram

SPRINT 2



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SPRINT 2



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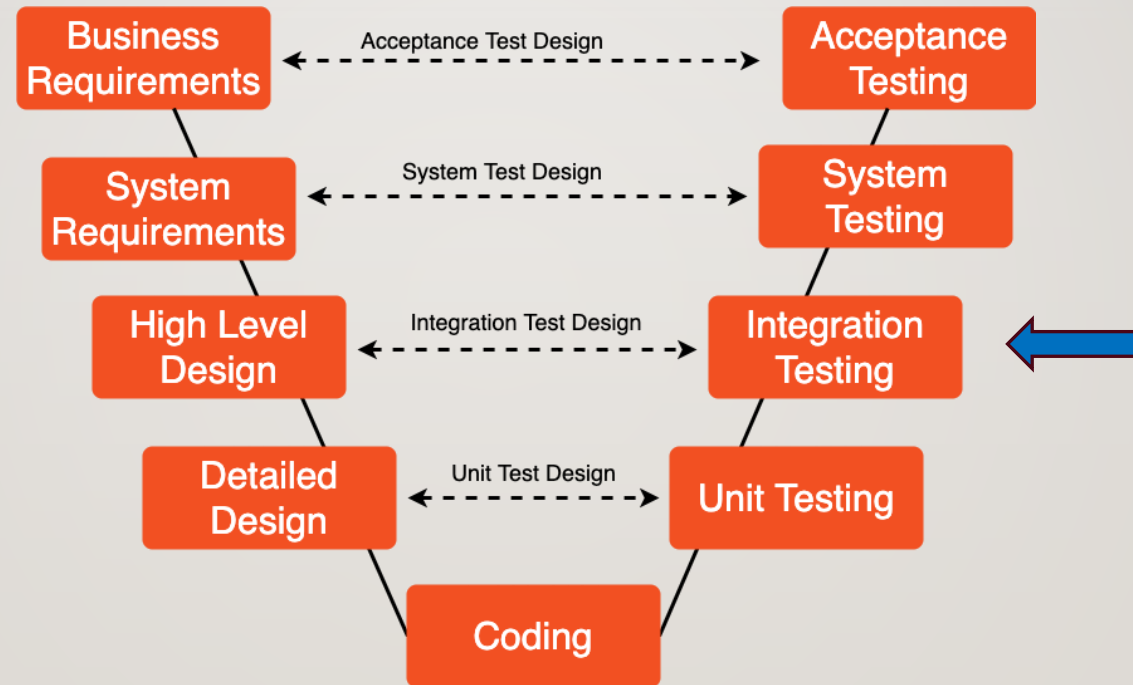
TEAM EXERCISE 15

Perform coding and testing of Sprint 2 Module

(평가 3) DESIGN/CODE/TEST REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

SPRINT 1&2 – INTEGRATION & TEST



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TEAM EXERCISE 16

Perform integrate and test of Sprint 1&2 Modules

SETTING SOLUTION NETWORK

DAY 5 - System Monitor & Multi Robot

Aa 이름

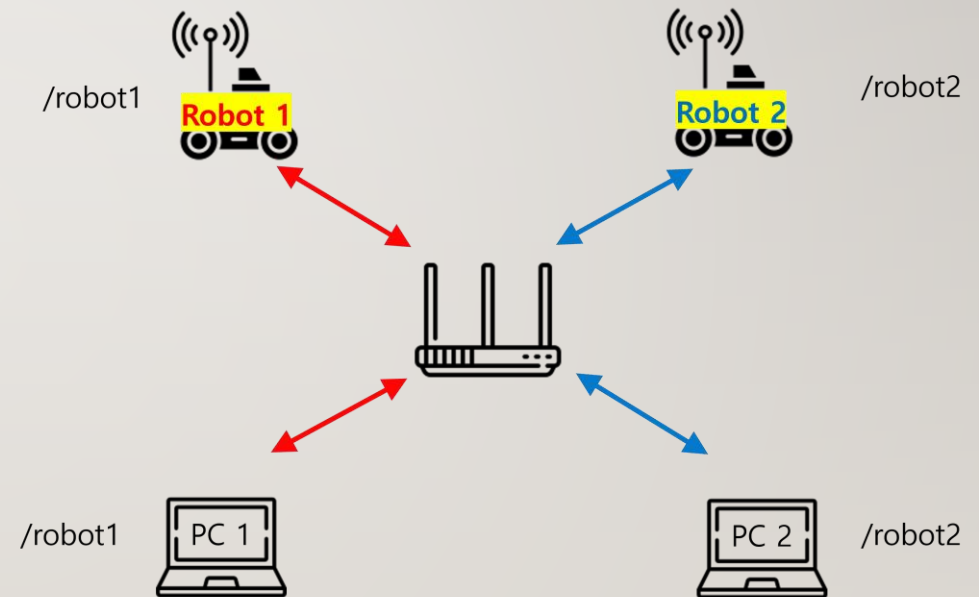
☀ 상태

● Multi Robot Standard Setup

● 시작 전

● Multi Robot Custom Discovery Setup

● 시작 전



프로젝트 RULE NUMBER ONE!!!

Are we having
Fun???

